BIRATIONAL MORPHISMS OF DECENT SPACES ARE ISOMORPHIC OVER A DENSE OPEN

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5301-5306, immediately following the statement at 5287-5299.

Editorial note. The two 'may assume' reductions are retained verbatim; the equivariance reading recorded in proof_roadmap is the reviewer's checklist, not author text.

Editorial note. One cross-chapter pointer (Morphisms) and two same-chapter line-indexed stubs are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Reduction to a scheme target. Lemma lemma-finitely-many-irreducible-components of this chapter, cited at proof lines 5301-5302. May assume Y is a scheme (maximal representable open, finitely many irreducible components).
- Reduction to a finite morphism. Lemma lemma-finite-over-dense-open of this chapter, cited at proof lines 5303-5304. May assume f is finite over a dense open; then X is a scheme too.
- Scheme-side birational isomorphism over a dense open. Morphisms, Lemma morphisms-lemma-birational-isomorphism-over-dense-open, cited at proof lines 5305-5306. The imported terminal isomorphism for finite birational morphisms of schemes.

Source reference. Original lemma 0BBC: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 5111–5111. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BB9: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4911–4911. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BAJ: focused-seeds/geometry/S9-decent-birational-isomorphism-over-dense-open/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a birational morphism of algebraic spaces over S which are decent and have finitely many irreducible components. Assume*

(1) *either f is quasi-compact or f is separated, and*
(2) *either f is locally of finite type and Y is reduced or f is locally of finite presentation.*

Then there exists a dense open $V \subset Y$ such that $f^{-1}(V) \to V$ is an isomorphism.

Proof. By Lemma 0BB9 we may assume Y is a scheme. By Lemma 0BBC we may assume that f is finite. Then X is a scheme too and the result follows from Morphisms, Lemma 0BAJ. □